Lecture 05: Workshop - Data Modeling

ITCS216 Fundamentals of Database Systems
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RECAP: Developing ERD

- Iterative process
- Usually involve the following activities:
    1. Create a detailed narrative of the organization's description of operation
    2. Identify the business rules based on the description of operation
    3. Identify the main entities and relationships from the business rules
    4. Develop the initial ERD/EERD
    5. Identify attribuites, keys
    6. Revise and review

Develop an ERD Guideline

- Key questions
  - What are the entities invoved?
  - What is the relationship?
  - What is the **connectivity** constraint of relationship? 1-M, M-N, or 1-1
  - What is the **cardinality** constraint of relationship? min. and max
  - What is the **participation** constraint of the relationship? total vs. optional
- Other details
  - Strong or Weak entity?
  - Strong or weak relationship?
  - Keys and attributes (regular / mu;ti-value / composites)

Develop an EERD Guideline

- Specializaiton
  - Supertype – subtypes
  - What are their common atrributes for a “supertype”?
  - What are the unique attributes for each “subtype”?
  - Specify subtype discriminator and values
  - Discrimiator constraints
    - Disjoint vs. Overlapping
    - Total vs. Partial

Part 1 (1.5 hr) – Refine the Biz Rules

- Use the **group project** to do an **individual** assignment
- Each member picks the different business process
- Revise or create at least **3 business rules** using **Template#1**
- Revise or create at least **2 business rules** using **Template#2**

Templates for Biz rules

Template#1: Format: Noun + Verb + Noun

Business Process Name:		
Rule No. #	**Rule Statement**	**Relationship**
Forward/Active		
Backward/Passive		
Conclusion		

Template#2: Format: Noun + Verb + Attributes

Business Process Name:
Rule No. #

Part 2 (1 hr) – Biz to EERD

- Use the result from Part 1 to to draw the ERD (CHEN notation) using **Template #3**
- Use the result from your teammate to create only one EERD diagram per group and show the EERD. (Recommended Tools: https://www.drawio.com/)

Templates for EERD

Template#3

Business Process Name:	
Rule No. 1	
Rule No. …	
Rule No. X	
ERD	

Example: Template#1

Template#1: **Format:** Noun + Verb + Noun (e.g., a student enrolls many classes)

Business Process Name: Book Order		
Rule No. # 1	**Rule Statement**	**Relationship**
Forward/Active	A book is written by many authors	1:M
Backward/Passive	One author can write many books	1:M
Conclusion	Books are written by many authors	M:N

Example: Template#2

Template#2: **Format:** Noun + Verb + Attributes (e.g., a student comprises of ID, Name, Email),

Business Process Name: Book Management
Rule No. # 2
A book comprises of ISBN, book title, number of pages, and synopsis.

Example: Rule 1 and Rule2

A book comprises of ISBN, boot title, number of pages, and synopsis.

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